

Literal Source Attachment in the Four-Packet TPC Interface

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Abstract

The typed L^2 interface in the preceding stage leaves a source-identification question: does the actual comparison-weight readout see the projected output of the literal prime-shell operator? This note answers the question on the registered finite schedule without replacing the source by a model functional. For each of six growing-cutoff scales and both recorded kernel exponents, we recompute the literal V59 operator, the three-contrast block projection, the source profile, and the projected output. Outward interval arithmetic certifies that the actual scalar attachment $C = \langle w_\perp, S \rangle$ is separated from zero on all twelve rows: eleven are negative and one is positive. The smallest normalized attachment coefficient $C^2/(\|w_\perp\|^2\|S\|^2)$ is only about 3.36×10^{-5} . Thus the finite source lock is real but weak; it supplies neither a uniform asymptotic nondegeneracy theorem nor fixed-power credit. An independent exact replay and hostile mutation audit are included.

1 Question and scope

The four-packet analysis records a source vector β , a literal finite operator A , and a block projection P_3 . The output used by the signed reassembly is

$$S = (I - P_3)A\beta.$$

The arithmetic readout is not an arbitrary dual vector. It is the actual comparison-weight vector w projected onto the same transverse space:

$$w_\perp = (I - P_3)w, \quad C = \langle w_\perp, S \rangle.$$

The purpose of this paper is to audit this scalar on the frozen finite source chain. It is not to infer a power law from a finite table. The distinction is important because the typed Gate-B interface of TPC-281 requires an additional source-identification or nondegeneracy statement; the present computation tests the literal object that such a statement would have to control. The packet and typed-interface inputs are the registered artifacts of TPC-275 and TPC-281 [1, 2].

The physical matrix has entries, for distinct indices u, t ,

$$A(u, t) = \sum_{q \in \mathcal{Q}} q K_H(u - t) \mathbf{1}_{q^t u} \mathbf{1}_{q^t t} \left(\mathbf{1}_{u \equiv t \pmod{q}} - \frac{1}{q - 1} \right), \quad (1)$$

with $K_H(h) = H^{2s}(H^2 + h^2)^{-s}$. The six triples (X, H, Q) are

$$(64, 15, 4), (96, 20, 5), (128, 24, 5), (192, 32, 6), (256, 38, 6), (384, 50, 7),$$

and the comparison cutoff is the registered growing cutoff from the upstream source certificate. The source interval is the outward interval for the literal prime-power weight minus the finite comparison profile.

2 Projection formula and attachment coefficient

Let the index interval be divided into four consecutive blocks of size b . The three contrast vectors are

$$a_1 = (1, 1, -1, -1), \quad a_2 = (1, -1, 0, 0), \quad a_3 = (0, 0, 1, -1),$$

with denominators $4b, 2b, 2b$, respectively. For a block-summed vector z , the projection energy is the sum of squared contrast sums divided by these denominators. Orthogonality therefore gives the following exact identity.

Proposition 1 (source-lock identity). *For the literal output $g = A\beta$ and source weight w ,*

$$C = \langle w, g \rangle - \langle P_3 w, P_3 g \rangle = \langle (I - P_3)w, (I - P_3)g \rangle.$$

If $W = \|w_\perp\|^2 > 0$ and $Y = \|S\|^2 > 0$, then the normalized attachment coefficient is

$$\rho^2 = \frac{C^2}{WY}, \quad 0 \leq \rho^2 \leq 1.$$

Proof. The three declared contrasts are mutually orthogonal after block expansion. Writing the orthogonal projection as the sum of the three rank-one pieces gives the first equality. The second follows from $S = (I - P_3)g$ and $I - P_3$ being self-adjoint and idempotent. The final inequality is Cauchy–Schwarz. $\square$

The certificate evaluates C and W with outward intervals. The matrix, β , and S are rational, so Y is exact rational arithmetic. A row is called *sign-separated* when the interval for C lies strictly on one side of zero. This is a finite, reproducible predicate rather than an asymptotic assertion.

3 Finite source-lock result

Theorem 2 (registered literal source attachment). *On the twelve registered rows (X, H, Q, s) , every interval for C is sign-separated, every W and Y is positive, and every interval for ρ^2 lies strictly inside $(0, 1)$. The sign census is eleven negative rows and one positive row. The smallest lower endpoint of ρ^2 occurs at $(X, H, Q, s) = (256, 38, 6, 2)$ and is*

$$\frac{167799905983}{5000000000000000} \approx 3.357 \times 10^{-5}.$$

Proof. The executable certificate uses the exact operator in (1), the locked source intervals, and the three projection contrasts. Its independent replay reconstructs the same quantities without importing the producer. The twelve resulting intervals are listed in Table 1; their endpoints separate zero and the lower endpoint of each ρ^2 is positive. Comparing the twelve rational lower endpoints gives the stated minimum. The source code also binds each row to the exact signed output energy in the TPC-275 certificate. $\square$

The row at $X = 256$, exponent 2, is not a typo: the finite source attachment changes sign there. A sign change does not invalidate the finite nonzero certificate, but it prevents a silent convention that the literal readout has one fixed phase on the entire registered schedule.

Table 1: Outward intervals for the actual attachment and its normalized square. The decimal display is shortened; the certificate stores the displayed endpoints as exact rational values.

X	H	Q	s	C interval	ρ^2 interval
64	15	4	1	[-27.4261,-27.4250]	[.0748099,.0748212]
64	15	4	2	[-21.8964,-21.8950]	[.0614523,.0614634]
96	20	5	1	[-53.6663,-53.6642]	[.0898041,.0898158]
96	20	5	2	[-45.3984,-45.3966]	[.0833289,.0833401]
128	24	5	1	[-31.1443,-31.1416]	[.0155791,.0155829]
128	24	5	2	[-35.9357,-35.9329]	[.0251363,.0251419]
192	32	6	1	[-7.01681,-7.00582]	[.00005266,.00005284]
192	32	6	2	[-33.1370,-33.1266]	[.0015391,.0015402]
256	38	6	1	[-37.2142,-37.2017]	[.0011437,.0011446]
256	38	6	2	[5.43945,5.45064]	[.00003357,.00003372]
384	50	7	1	[-91.8630,-91.8326]	[.0009659,.0009667]
384	50	7	2	[-43.6411,-43.6139]	[.0002702,.0002706]

4 What the result does and does not buy

The finite result resolves one ambiguity left by an abstract equal-norm functional: the actual source readout is genuinely present on each tested row. It does not resolve the uniform question. In particular, the minimum of ρ^2 is roughly four orders of magnitude below one, and the sign is not constant. No inequality of the form

$$|C_X| \geq cX^{-\theta} \quad \text{or} \quad \rho_X^2 \geq cX^{-\kappa}$$

is proved here. Such an inequality would have to be uniform on the same growing source, shell, height, and projection schedule.

The distinction also matters for TPC-281. A typed operator estimate $\|A_X\|_{2 \rightarrow 2} \leq KX^{-\sigma}$ controls an output from above, while the scalar C identifies one particular source direction. The present certificate supplies finite evidence for that identification, but neither the operator estimate nor a power-saving lower bound for the attachment. Therefore the arithmetic L^2 gate, fixed-power credit, and full Gate B remain open.

5 Reproducibility and claim firewall

The producer hash-locks the TPC-275 producer and canonical result, as well as the TPC-268 source engine that defines the comparison intervals. It reruns all twelve rows and writes canonical sorted-key JSON. The independent checker reconstructs the matrix output, block projection, scalar, source norm, exact output norm, and cosine interval directly. A hostile checker mutates an endpoint, sign, cosine interval, provenance hash, budget flag, and row census; all six mutations are rejected. Normal and optimized Python executions are required to have empty standard error and byte-identical output.

These checks establish a finite numerical certificate. They do not turn a finite source scan into a theorem uniform in X , and they do not claim a twin-prime conclusion.

6 Conclusion

The literal source attachment is now source-locked on the complete registered TPC-275 schedule: it is nonzero on twelve rows, with eleven negative and one positive signs. The strongest obstruction is quantitative rather than binary: the weakest normalized attachment is small. The next natural problem is to compute the exact perturbation radius at which this attachment can vanish and to test whether the observed weak margin is stable under nearby source choices.

References

- [1] Liang Wang. Signed four-packet reassembly on the literal v59 output. TPC-275 in-repository research artifact, 2026.
- [2] Liang Wang. A typed arithmetic l2 interface for four-packet gate-b reassembly. TPC-281 in-repository research artifact, 2026.